SSL4SAR: Self-Supervised Learning for Glacier Calving Front Extraction From SAR Imagery

Nora Gourmelon, Marcel Dreier, Martin Mayr, Thorsten Seehaus, Dakota Pyles, Matthias Braun, Andreas Maier, *Senior Member, IEEE*, and Vincent Christlein

***Abstract*— Glaciers are losing ice mass at unprecedented rates, increasing the need for accurate, year-round monitoring to understand frontal ablation, particularly the factors driving the calving process. Deep learning models can extract calving front positions from synthetic aperture radar (SAR) imagery to track seasonal ice losses at the calving fronts of marine- and lake-terminating glaciers. The current state-of-the-art model relies on ImageNet-pretrained weights. However, they are suboptimal due to the domain shift between the natural images in ImageNet and the specialized characteristics of remote sensing imagery, in particular for SAR imagery. To address this challenge, we propose two novel self-supervised multimodal pretraining techniques that leverage SSL4SAR, a new unlabeled dataset comprising 9563 Sentinel-1 (S1) and 14 Sentinel-2 (S2) images of Arctic glaciers, with one optical image per glacier in the dataset. Additionally, we introduce a novel hybrid model architecture that combines a Swin-Transformer encoder with a residual convolutional neural network (CNN) decoder. When pretrained on SSL4SAR, this model achieves a mean distance error (MDE) of 293 m on the "CAlving Fronts and where to Find thEm" (CaFFe) benchmark dataset, outperforming the prior best model by 67 m. Evaluating an ensemble of the proposed model on a multi-annotator study of the benchmark dataset reveals an MDE of 75 m, approaching the human performance of 38 m. This advancement enables precise monitoring of seasonal changes in glacier calving fronts.**

***Index Terms*— Deep learning, glacier calving fronts, multimodal, self-supervision, transformer, unsupervised pretraining.**

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I. Introduction

GLACIERS are facing record ice losses, with the largest annual mass loss rate ever recorded at 602 ± 69 Gt in 2023, and the trend is upward, as global glacier mass loss has been increasing exponentially in recent years [1]. For marine-terminating glaciers, which account for 40% of total glacier area [2], ice depletion is controlled by two mechanisms: negative surface mass balance and ice losses at the calving front [3], [4]. Glacier responses to ocean forcing and ice–ocean interaction are to date not well-understood [5]. To identify drivers of frontal ablation, and thus mass loss at the terminus, it is necessary to assess calving front variations. The community lacks large spatiotemporal databases of calving front positions that can be used as a basis for further analysis. Due to the impracticality of manually delineating calving fronts in satellite imagery, several attempts have been made in recent years to automate the extraction of front positions using deep learning (e.g., [6], [7]). Since seasonal calving front dynamics are of interest, and polar night and weather conditions limit the availability of optical images, we focus on synthetic aperture radar (SAR) images in this study. However, SAR images are more difficult to interpret than optical images due to speckle noise, often lower resolution, and similar backscatter patterns for different ice types [8]. On the other hand, the contrast between the open ocean and glacier ice is high for SAR images, particularly when ice mélange and sea ice are absent at the glacier front. For calving front delineation, a benchmark dataset with only SAR imagery exists: "CAlving Fronts and where to Find thEm" (CaFFe) [9]. A comparison of previously published deep learning models for calving front and coastline extraction on this benchmark [10] shows that the best performing model is based on a Swin-Transformer-based architecture without convolutions, denoted as "HookFormer" [11]. The model addresses the perennial challenge for deep learning models in remote sensing: how to incorporate surrounding contextual information into the local patch, with two U-shaped branches, where one receives the local patch and the other receives a surrounding, larger, downscaled patch. A drawback of this method is that it produces jagged calving front delineations. Therefore, this study proposes an optimized, hybrid transformer-convolutional neural network (CNN) architecture that reduces model size by eliminating the two-branch system and produces smooth calving front delineations through its CNN decoder.

HookFormer is currently the only model for calving front delineation that relies on pretrained weights

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rather than random initialization. Specifically, it uses Swin-Transformer [12] weights trained on the ImageNet dataset [13], which are publicly available.[1] This pretraining was crucial for achieving state-of-the-art performance and highlights the importance of using pretrained models (see Table I). ImageNet consists of natural images that have specific properties, which distinguish them from remote sensing images, such as perspective, scale, and spatial resolution. In addition, the images in ImageNet are optical images, unlike the SAR images in CaFFe. Such a discrepancy in the domain between the source dataset used for pretraining and the target dataset used for fine-tuning can affect the performance of the model in the downstream task [14], [15], [16], [17], [18]. To leverage and further expand the performance gains of pretraining for calving front delineation, we introduce a new unlabeled SAR dataset of calving glaciers together with two novel in-domain, multimodal pretraining strategies that use SAR images as input and optical images as labels.

Finally, we present a new state-of-the-art deep learning ensemble model for calving front delineation in SAR images, which achieves near-human performance in a comparison with the multiannotator study on the CaFFe benchmark [10]. Our contributions can be summarized as follows:

1) introduction of an SAR dataset of arctic glaciers for self-supervised pretraining;
2) two novel multimodal pretraining strategies;
3) proposal of an optimized single-branch model architecture;
4) new state-of-the-art ensemble model for calving front delineation in SAR imagery with near human performance.

The remainder of this article is organized as follows. Section II gives an overview of the related work. Section III describes the datasets used. The proposed architecture and pretraining strategies are explained in Section IV, while Section V outlines the evaluation methods, training setup, and ensemble approach. The results are presented in Section VI, followed by an ablation study on the ensemble approach components in Section VI-G. Finally, Sections VI-H and VII provide a discussion of the findings and conclude this article, respectively.

II. Related Work

Several studies have proposed deep learning models for extracting calving front or coastline positions from satellite imagery. So far, only one of these studies has used supervised pretraining, and none have performed in-domain pretraining. However, there is a large body of literature on self-supervised pretraining for remote sensing. In Sections II-A and II-B, we review studies on delineating calving fronts and coastlines in satellite imagery and studies on self-supervised pretraining for remote sensing.

A. Calving Front Delineation

Classical calving front extraction approaches [19], [20], [21], [22], [23] rely on either abrupt intensity changes at the terminus, relatively homogeneous neighboring glacier ice and ocean areas with different average intensities, or both [19]. However, these assumptions often do not hold in complex SAR scenes, limiting the generalizability of such methods. To address this, recent research has increasingly adopted data-driven techniques. Deep learning models for delineating calving fronts or coastlines from satellite imagery can be broadly categorized into CNN-based and transformer-based architectures.

The majority of studies rely on CNNs [6], [7], [9], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. Since the receptive field of CNNs is locally bounded, techniques are needed to introduce global information. Widely used are encoder-decoder architectures such as the U-Net [40] and DeepLabv3+ [41], which use multiple pooling operations to aggregate information from more distant locations [7], [9], [24], [25], [26], [27], [28], [29], [30], [31], [32], [34], [35], [36], [37], [38], [39]. Deeper architectures [7], i.e., with more pooling operations and larger kernel sizes [37], are sometimes used to enhance the aggregation effect of encoder–decoder structures. To further increase the receptive field, several studies use atrous spatial pyramid pooling (ASPP) [9], [25], [28], [35], [38], [39], [42] and dilated (atrous) convolutions in general [35]. Two studies [30], [31] apply deep supervision to reinforce the global information aggregated by their encoder–decoder architectures. In postprocessing, Gourmelon et al. [28] incorporate a conditional random field (CRF), which, unlike simple thresholding, considers the potentially assigned classes of all other pixels when determining the class of a given pixel based on the neural network's output. Finally, Wu et al. [36] implement a two-branch model that receives a local and a context patch. Information is transferred from the context branch to the local branch, effectively increasing the field of view.

Transformer-based networks, on the other hand, have access to global information due to the inherent attention mechanism. The successor model [11] of Wu et al.'s [36] two-branch model still has two branches, but replaces the convolutions with Swin-Transformer [12] blocks. Zhu et al. [42] combine transformers and CNNs by incorporating global-local attention Swin-Transformer blocks in DeepLabv3+.

B. Self-Supervised Pretraining for Remote Sensing

With the increasing availability of ever-larger unlabeled datasets, self-supervised pretraining is on the rise and has not stopped at remote sensing.

Contrastive learning is a widely used self-supervised learning (SSL) technique that reduces the distance between semantically identical samples (positive pairs) in feature space and, for some approaches, also increases the distance between samples that are not semantically identical (negative pairs) [43]. In remote sensing, common ways to generate positive pairs include flipping [15], [17], [18], [44], [45], [46], rotating [18], cropping [15], [16], [17], [18], [44], [45], [46], [47], resizing [15], [46], masking [15], [45], [46], shifting [48], blurring [16], [18], [44], [47], color jitter [16],

[1] https://github.com/microsoft/Swin-Transformer

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TABLE I

MDEs IN METERS FOR CAFFE'S TEST SET DIVIDED BY CAPTURING SENSOR. A BOLD FONT SIGNIFIES THE BEST VALUE WITHIN EACH DIVISION OF A COLUMN. WEIGHTS ARE THE WEIGHTS AT INITIALIZATION OF THE MODEL. SETUP REFERS TO THE CONFIGURATION OF THE TRAINING PIPELINE. PALSAR IS SHORT FOR ALOS PHASED ARRAY L-BAND SYNTHETIC APERTURE RADAR. TSX STANDS FOR TerraSAR-X AND TanDEM-X

	Architecture	*pretraining*	*Weights*	*Setup*	*All*	*Sentinel-1* ↓ *MDE*	*ENVISAT* ↓ *MDE*	*ERS* ↓ *MDE*	*PALSAR* ↓ *MDE*	*TSX* ↓ *MDE*
	HookFormer	/	Scratch	1	938 ± 90	2654 ± 277	1004 ± 364	983 ± 610	678 ± 208	760 ± 64
		/	ImageNet	2	360 ± 13	918 ± 76	253 ± 42	174 ± 47	263 ± 28	286 ± 8
	TYRION	/	Scratch	1	682 ± 101	1604 ± 230	644 ± 148	512 ± 520	346 ± 114	550 ± 97
		/	ImageNet	2	342 ± 25	750 ± 79	325 ± 96	168 ± 50	184 ± 45	274 ± 24
		OptSimMIM	Scratch	3	773 ± 92	1961 ± 407	598 ± 264	448 ± 357	364 ± 161	610 ± 76
		OptSimMIM	ImageNet	4	300 ± 60	$\mathbf{592 \pm 175}$	307 ± 123	154 ± 68	192 ± 65	253 ± 42
		OptTranslator	Scratch	3	574 ± 92	1364 ± 175	629 ± 148	485 ± 358	338 ± 109	457 ± 90
		OptTranslator	ImageNet	4	$\mathbf{293 \pm 54}$	638 ± 309	$\mathbf{234 \pm 67}$	$\mathbf{148 \pm 68}$	$\mathbf{182 \pm 78}$	$\mathbf{240 \pm 14}$
Ensem.	TYRION	/	ImageNet	2	308	674	587	137	200	243
Ensem.	TYRION	OptSimMIM	ImageNet	4	**238**	**398**	456	**70**	**142**	**209**
Ensem.	TYRION	OptTranslator	ImageNet	4	246	425	**296**	**70**	147	217
					Multi-Annotator Study					
Ensem.	HookFormer	/	ImageNet	2	221 ± 15	915 ± 108	230 ± 40	157 ± 46	186 ± 32	105 ± 5
Ensem.		/	ImageNet	2	131	437	591	108	161	67
Ensem.	TYRION	OptSimMIM	ImageNet	4	86	241	476	47	104	49
Ensem.		OptTranslator	ImageNet	4	75	254	285	53	**100**	38
Ensem.	Humans				$\mathbf{38 \pm 15}$	$\mathbf{99 \pm 26}$	$\mathbf{151 \pm 228}$	$\mathbf{22 \pm 6}$	129 ± 182	$\mathbf{21 \pm 6}$

[17], [47], grayscale shifting [47], solarization [16], [47], Gaussian noise [43], and, in the case of multimodal models, sensor dropping [16], and any combination of the above. If a multitemporal dataset is used, positive pairs can also be formed by taking temporally close images [17], [47], [49].

In contrast, generative SSL techniques typically reconstruct the input data or generate new data based on the input, often employing autoencoders or generative adversarial networks (GANs) [43]. A classic pretext task for generative SSL is masked image modeling (MIM) [50], [51], which involves masking parts of the input with zeros and subsequently reconstructing them using the model. However, besides MIM [52], pretext tasks such as denoising [53] or band switching [54] are also used for pretraining in remote sensing.

Most remote sensing studies rely primarily on optical imagery for pretraining. A work that takes SAR–optical image pairs as input is presented by Scheibenreif et al. [55]. The approach uses a model for each modality and applies a contrastive loss to bring the SAR-extracted and optical-extracted features closer together. Similarly, Chen and Bruzzone [48] and Wang et al. [16] take SAR–optical image pairs as input, but only employ one shared model with the modalities in separate channels. In [48], both the pretraining and the fine-tuning use multimodal data, whereas Wang et al. [16] show that multimodal pretraining helps downstream tasks with multimodal, only SAR, and only optical data. A work that takes only SAR images as input is presented by Li et al. [18]. However, their contrastive approach still requires SAR–optical image pairs. To generate a positive sample for the real SAR image, they employ a GAN to create a fake SAR image based on the optical image.

In contrast, our multimodal, generative pretraining technique does not necessitate temporally aligned SAR–optical image pairs and manages with only a limited number of geographically co-registered optical images.

III. DATASETS

This work introduces novel multimodal pretraining approaches that make use of new unlabeled data. For fine-tuning and evaluation, we use the manually labeled CaFFe benchmark dataset [56].

A. SSL4SAR

The unlabeled dataset for pretraining, which we call SSL4SAR, comprises 9562 Sentinel-1 (S1) images in interferometric wide swath mode covering the period 2015-2022, depicting 14 distinct glaciers in the Arctic with different sizes and calving front geometries. A bounding box around each glacier terminus was manually defined. In order to efficiently obtain time series of SAR imagery for the target glacier, we relied on S1 SAR ground range-detected (GRD) image collection [ee.ImageCollection("COPERNICUS/S1_GRD")] provided by the database from Google Earth Engine (GEE). In contrast to the original S1 SAR GRD imagery provided by Copernicus, GEE applied additional preprocessing steps to enhance the quality of the data: 1) apply orbit files; 2) GRD broader noise removal; 3) thermal noise removal; 4) radiometric calibration; and 5) terrain correction. At each target glacier, the S1 imagery was subsetted to the defined bounding box and reprojected to Arctic polar stereographic projection (EPSG: 3995) for the glaciers on Svalbard and UTM zone 6N projection (EPSG:32606) for the Columbia glacier at 10 m spatial resolution.

In addition to the SAR imagery, for each of these glaciers, one cloud-free, optical image captured by Sentinel-2 (S2) between June and August 2020 is included in the dataset. The images were manually selected from the GEE S2 MSI

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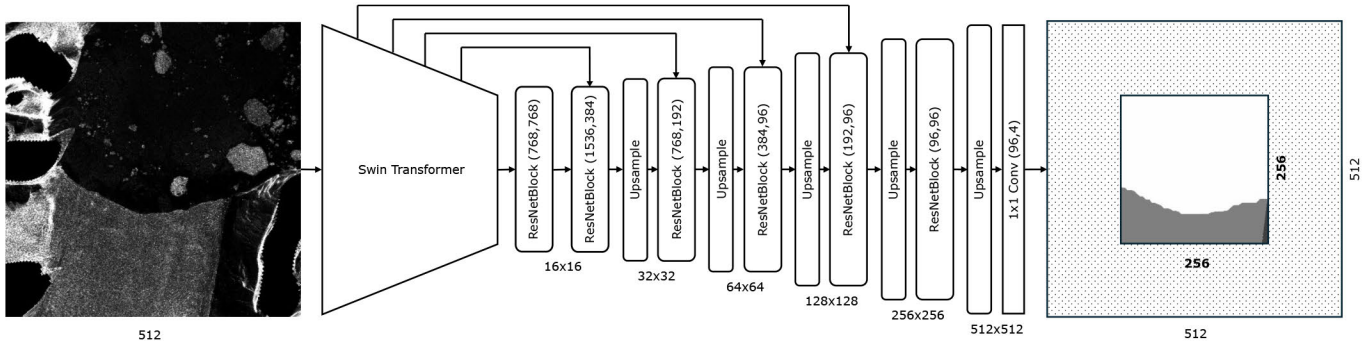


Fig. 1. Architecture of the network, dubbed "TYRION" (Transformer Hybrid with a Residual Convolutional Neural Network). The Swin Transformer requires three input channels, so the single-channel SAR image is replicated three times before being input into the model. The output consists of four channels, each representing a class: ocean (white), glacier (light gray), rock outcrop (dark gray), and areas with no information. The outer frame of the output is not considered during testing and inference but included during training. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model.

collection [ee.ImageCollection("COPERNICUS/S2_SR")]. Similar to the S1 imagery, the multispectral data were cropped to the bounding boxes defined at each glacier terminus and reprojected to 10-m spatial resolution at the respective projections used for the SAR imagery at each glacier. The multispectral images feature all 12 bands as well as the data products aerosol optical thickness (AOT), average water vapor (WVP), scene classification layer (SCL), true color image (TCI), cloud probability mask (MSK_CLDPRB), snow probability mask (MSK_SNWPRB), and cloud masks (QA10, QA20, and QA60). However, we only utilize the 12 bands, WVP, and SCL in our pretraining approaches. WVP and SCL offer additional informative supervision signals, providing semantically meaningful spatial structures that can be visually interpreted and are correlated with surface characteristics relevant to the downstream task of landscape zone segmentation.

B. CaFFe

CaFFe consists of 681 SAR images acquired by six different sensors with varying spatial resolutions, one of which is S1. The dataset includes scenes captured from both ascending and descending orbits, resulting in a range of viewing geometries and incidence angles. The acquisition period runs from 1996 to 2020. The dataset depicts seven marine-terminating glaciers, five situated on the Antarctic Peninsula, one in Greenland, and one in Alaska. Following the official train–test split, all images from two of these glaciers—Mapple and Columbia glaciers—are assigned to the test set, resulting in a total of 122 test images. CaFFe provides two labels for each image. One label shows the calving front of the glacier, while the other label, which is intended as an intermediate objective for segmentation approaches, depicts landscape zones (ocean, glacier, rock, and no information available). For further information on CaFFe, we refer the reader to [9]. To assess the reliability of the manual annotations and establish a human performance reference for automated calving front extraction, Gourmelon et al. [10] quantified the inter-annotator variability by computing the mean distance error (MDE) between calving front delineations provided by ten independent annotators across all images in the test set.

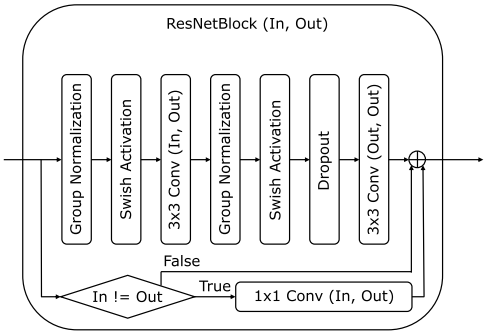


Fig. 2. Design of the ResNetBlock. ⊕ indicates addition.

IV. Methodology

A major challenge for deep learning models in remote sensing is the integration of context information into the model. Previous studies [11], [36] for calving front delineation in SAR imagery have used dual-branch architectures. With our new hybrid transformer-CNN model, we eliminate the need for a second branch.

A. Architecture

We take inspiration from the current best performing method, the fully transformer-based model "HookFormer" [11]. However, we simplify its architecture and overcome the problem of jagged fronts produced by the original model. The model receives a single-channel SAR image, replicated three times to match the encoder's input channel count, and outputs four segmentation maps—one for each landscape zone. The model's architecture is illustrated in Fig. 1. The encoder–decoder structure is a hybrid between a transformer and a CNN. The network's encoder is a SwinV2-Transformer [57] with four downsampling steps and window size of 16, while the CNN decoder consists of residual blocks [58] and upsampling steps. The design of the residual block, taken from [59], is depicted in Fig. 2, and the upsampling is performed by nearest neighbor interpolation. The CNN decoder produces zone segmentations that result in smooth fronts after postprocessing. To eliminate the need of a second branch, we increase the input size of the model from 224×224 to 512×512. The input size is maintained by

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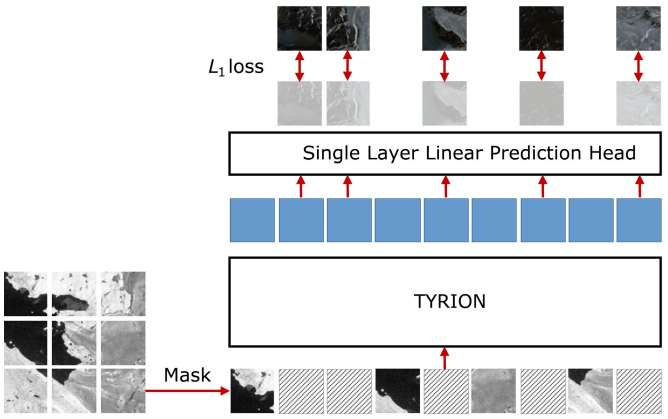


Fig. 3. Visualization of SSL4SAR–OptSimMIM. For illustration purposes, we show only the rgb channels of the optical image. The striped patches are overwritten with mask tokens consisting of zeros. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model. Figure is adapted from [51].

the network, but we retain only the inner 256 × 256 patch to ensure that a sufficient amount of surrounding context information is used for the output. We dub our network "TYRION" (Transformer Hybrid with a Residual Convolutional Neural Network).

B. SSL Pretraining

When humans delineate calving fronts on SAR imagery, they often use an additional optical image of the glacier as a reference to support the mapping. Deep learning models can also benefit from multimodal training, as useful and possibly complementary information is transferred from both modalities, thus improving the quality of latent spatial features [60], [61]. In line with the typical human approach, we have implemented two multimodal pretraining techniques: an Optical supervised SimMIM trained on SSL4SAR (SSL4SAR–OptSimMIM) and SAR to Optical Translator trained on SSL4SAR (SSL4SAR–OptTranslator). In contrast to previous studies [16], [18], [48], [55], both presented pretraining techniques use only one additional optical image per glacier, rather than a temporally matching optical image for each SAR image in the SSL4SAR dataset. This markedly reduces the need for manual data curation, as the number of cloud-free matching optical images to be collected is reduced from 9562 to 14 in the case of SSL4SAR.

1) SSL4SAR–OptSimMIM: The first pretraining approach, called SSL4SAR–OptSimMIM, leverages the simple framework for MIM (SimMIM) [51]. SimMIM belongs to the generative SSL methods. It masks random patches of the input image and uses a one-layer prediction head to predict the raw pixel values of these masked patches. SSL4SAR–OptSimMIM receives an SAR image and uses the one-layer prediction head to predict the values of the corresponding patches of the optical image from the same glacier instead of predicting the SAR patches. A visualization of SSL4SAR–OptSimMIM is given in Fig. 3.

2) SSL4SAR–OptTranslator: The second pretraining approach is a straightforward generative SSL method that employs TYRION with an additional single-layer convolutional head to translate each SAR image into its corresponding optical counterpart. SSL4SAR–OptTranslator is illustrated in Fig. 4.

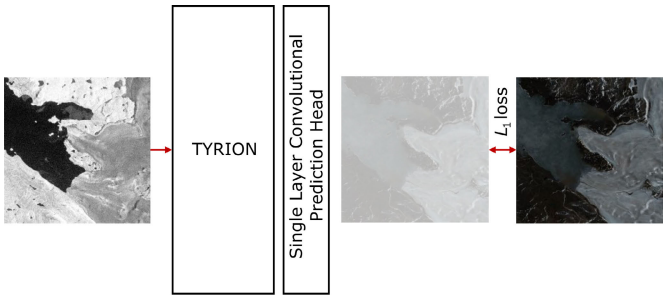


Fig. 4. Visualization of SSL4SAR–OptTranslator. For illustration purposes, we show only the rgb channels of the optical image. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model.

V. Experiments

To demonstrate the effectiveness of the two pretraining techniques and the TYRION architecture, we perform a series of experiments with and without pretraining and compare the results with those of the HookFormer [11], the current state-of-the-art model [10] for extracting calving fronts from SAR.

A. Evaluation Metrics

The primary evaluation metric for calving front delineation is the MDE [9], which gives the symmetric mean distance between the predicted and ground-truth calving front. The MDE [9] is calculated as

$$\mathrm{MDE}(\mathcal{I}) = \frac{1}{\sum_{(\mathcal{P},\mathcal{Q})\in\mathcal{I}}(|\mathcal{P}|+|\mathcal{Q}|)} \cdot \sum_{(\mathcal{P},\ \mathcal{Q})\in I}\left(\sum_{\vec{p}\in\mathcal{P}}\min_{\vec{q}\in\mathcal{Q}}\|\vec{p}-\vec{q}\|_2 + \sum_{\vec{q}\in\mathcal{Q}}\min_{\vec{p}\in\mathcal{P}}\|\vec{p}-\vec{q}\|_2\right) \tag{1}$$

where $\mathcal{P}$ is the set of ground-truth calving front pixels of an image, $\mathcal{Q}$ is the set of predicted calving front pixels of that image, and $\mathcal{I}$ is the set of all images. The cardinality of a set is represented as $|(\cdot)|$ in the equation.

B. Training Setup

A visualization of the training pipeline with the employed datasets can be found in Fig. 5. The pipeline consists of three stages. In stage 1, the model encoder is initialized with ImageNet pretrained weights obtained through supervised learning; if stage 1 is excluded, the encoder is trained from scratch.

Stage 2 involves pretraining TYRION on SSL4SAR using either SSL4SAR–OptSimMIM or SSL4SAR–OptTranslator. For pretraining (stage 2), the SSL4SAR dataset is split into a training set and a validation set. The latter comprises the initial 68 images in the time series of each of the 14 glaciers, representing approximately 10% of the total dataset. Random resized crops and flipping are applied on the fly to augment the images. An L_1 loss function is applied.

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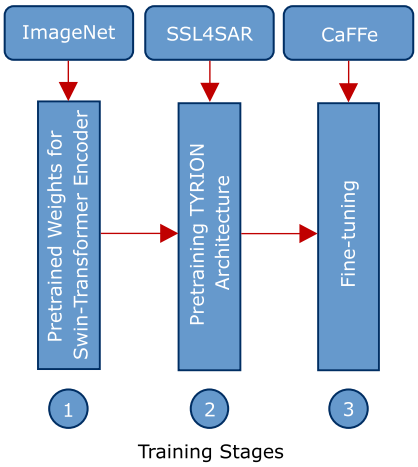


Fig. 5. Visualization of the training pipeline. In stage 1, TYRION's Swin-Transformer encoder is initialized with ImageNet pretrained weights. Stage 2 involves pretraining TYRION on SSL4SAR. In stage 3, TYRION is trained to segment landscape zones based on CaFFe's zone labels.

In stage 3, TYRION is fine-tuned to segment landscape zones based on CaFFe's zone labels using the train–test split defined by Gourmelon et al. [9]. The weights for fine-tuning are initialized with the pretraining checkpoint from the epoch with the lowest L_1 loss on the validation set. The applied loss function is a combination of the cross-entropy on the full model output, the cross-entropy on the inner 256×256 of the output, the dice score on the full model output, and the dice score on the inner 256×256 of the output. We apply label smoothing ($\epsilon = 0.1$) for the cross-entropy losses. Augmentations for fine-tuning consist of random resized crops, vertical and horizontal flips, rotations, brightness adjustments, gamma corrections, additive Poisson noise, and a modified mix-up variant that only alters the ocean area. For hyperparameter estimation, CaFFe's train set is further split into a 10% validation set and a 90% training set, while CaFFe's test set remains untouched until the final evaluation. For testing, the checkpoint with the lowest MDE on the validation set is chosen. Ultimately, the calving fronts are extracted from the segmented zones via the postprocessing pipeline outlined by Gourmelon et al. [9]. This pipeline involves several steps: first, patch merging and the argmax operation are applied. Next, a connected components analysis is performed on the predicted ocean zone to retain only the largest ocean region and fill in any potential gaps or islands. The boundaries between this ocean region and predicted glacier regions are then extracted as potential calving fronts. Additionally, bounding box masking is applied to focus on the region of interest, and any calving fronts shorter than 750 m are discarded.

We conduct experiments using four different setups to evaluate the performance of our pretraining strategy and the TYRION architecture, and to analyze the overall impact and importance of pretraining.

1) Setup 1 omits both Stages 1 and 2, serving as a baseline.
2) Setup 2 excludes only Stage 2 to assess the effect of domain-specific pretraining.
3) Setup 3 omits Stage 1 to investigate the role of weight initialization.
4) Setup 4 includes all three stages.

As a comparison to the TYRION architecture, the HookFormer [11] is tested in two configurations: one corresponding to Setup 1 and the other to Setup 2.

C. Ensemble

In order to achieve the optimal results during the inference stage, it is essential to apply the model in the most effective manner. Therefore, in addition to the experiments demonstrating the superiority of the presented novel architecture and pretraining approaches, we apply well-established test-time techniques to further improve the results. These techniques include a sliding window approach with an overlap (50%) for extracting patches from the test images, fourfold test-time augmentation through rotations, and the formation of an ensemble of five models by averaging their confidence values per image. The five models are retrained versions of TYRION that have gone through the entire training pipeline, starting with the ImageNet pretrained weights (setup 2 or 4). Furthermore, the ensemble enables the calculation of uncertainty maps, which are generated by computing the pixelwise standard deviation over the confidence values of the five models per class.

D. Multiannotator Evaluation

The MDEs on the test set are calculated by comparing the results with CaFFe's front labels for all experiments. However, Gourmelon et al. [10] have conducted a multiannotator study on CaFFe's test set, resulting in new averaged ground-truth front labels and an interannotator MDE. In order to assess how close our models (setups 2 and 4) are to human performance, we apply the additional postprocessing necessary for a fair comparison between humans and AI models, outlined by Gourmelon et al. [10], and compare the predictions with the new averaged front labels.

VI. Results

The quantitative results on the test set of the CaFFe dataset for all experiments are summarized in Table I, which also provides a sensor-specific breakdown of the results. The lowest average MDE of 293 m is achieved by TYRION SSL4SAR–OptTranslator initialized with ImageNet-pretrained weights (setup 4). Qualitative results are visualized in Fig. 6, which displays a sample image with its zone label and zone segmentations of the experiments, and Fig. 7, illustrating the predicted calving fronts for one sample image across TYRION (setup 2) and the two pretraining approaches—OptSimMIM and OptTranslator —(setup 4), along with their ensemble and postprocessed ensemble results for the multiannotator study.

A. Architecture Comparison: TYRION Versus HookFormer

Even without pretraining, TYRION (setup 2) outperforms the previous state of the art, the HookFormer [11] (setup 2), with a mean MDE improvement of 18 m. As intended with the CNN decoder, TYRION produces smoother calving front predictions compared to the HookFormer. As shown in Fig. 6, the HookFormer prediction exhibits jagged edges at the interface between the ocean and glacier, whereas the TYRION

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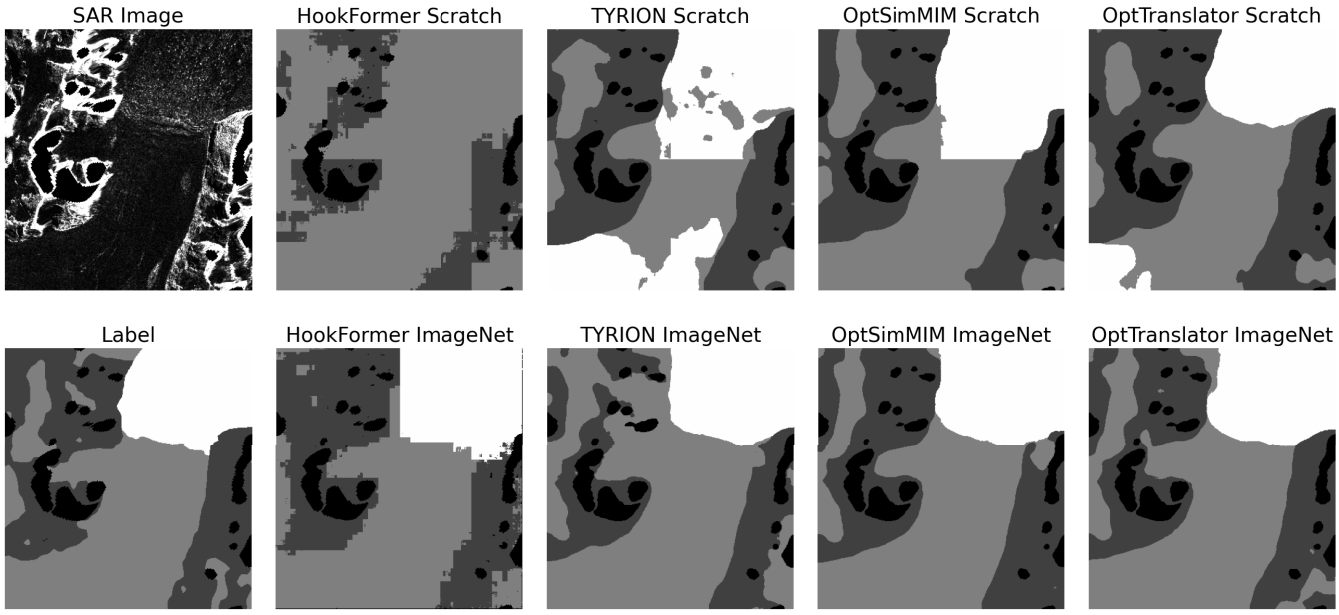


Fig. 6. Sample image of the Mapple glacier captured by S1 on January 8, 2020, with the corresponding zone label and zone segmentations of the experiments. The ocean is visualized in white, the glacier in light gray, the rock in dark gray, and black indicates areas where no information is available. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model.

output exhibits considerably smoother delineations. Additionally, the number of parameters has been reduced; while the HookFormer has 59.3G parameters, the TYRION architecture has only 50.9G parameters.

B. Impact of Pretraining

The integration of pretraining consistently improves TYRION's performance, irrespective of the SSL approach employed (setup 2 versus 4). For SSL4SAR–OptSimMIM, the MDE decreases by 42 m, while SSL4SAR–OptTranslator achieves a reduction of 49 m.

Qualitative improvements can be seen in Figs. 6 and 7. When comparing the pretrained TYRION models (setup 4) to HookFormer (setup 2), the MDE differences are larger, with 60- and 67 m reductions for SSL4SAR–OptSimMIM and SSL4SAR–OptTranslator, respectively. The observed difference between the two pretraining approaches is low with 7 m.

C. Role of Initialization: ImageNet-Pretrained Weights

Training the HookFormer from scratch (setup 1 versus 2) more than doubles its MDE and leads to an average of 25 images where no front is recognized at all out of the 122 images in the test set. Initialization with ImageNet-pretrained weights also enhances TYRION's performance (setup 3 versus 4 and setup 1 versus 2), both with and without the introduced SSL pretraining approaches. TYRION initialized with ImageNet-pretrained weights (setup 2) outperforms both TYRION trained from scratch (setup 1) and TYRION pretrained on SSL4SAR starting from scratch (setup 3). The benefits of initialization are evident in Fig. 6, where predictions initialized with ImageNet weights demonstrate a notable improvement in discerning between glacier and ocean regions.

D. Sensor-Specific Analysis

The impact of pretraining varies by sensor, as detailed in Table I. Pretraining with SSL4SAR–OptSimMIM (setup 2 versus 4) achieves the greatest MDE reduction for S1 images (21%), whereas changes for other sensors range between a reduction of 8% and a slight increase of 4%. In contrast, SSL4SAR–OptTranslator (setup 2 versus 4) yields the largest reduction for ENVISAT images (28%), followed by S1 images (15%). Overall, pretraining on SSL4SAR provides the greatest performance gains for S1 images.

E. Ensemble Approach

The ensemble approach, which leverages test-time augmentations and overlapping patches, yields further reductions in MDE. For TYRION without pretraining (setup 2), the ensemble approach decreases the MDE by 34 m, while reductions of 62 and 47 m are observed for SSL4SAR–OptSimMIM and SSL4SAR–OptTranslator (setup 4), respectively. Sensor-specific analysis reveals that ensemble approaches reduce the MDE for most sensors, except ENVISAT, where all three configurations—TYRION without pretraining and with the two pretraining approaches (setups 2 and 4)—show an increase in MDE. For PALSAR, only TYRION without pretraining (setup 2) results in an increase in MDE. Fig. 8 displays the uncertainty maps generated using the ensemble approach. Of particular interest is the uncertainty map for the ocean class, which effectively illustrates the model's confidence in delineating the ocean boundaries, with the exception of the left calving front, where there is a discernible degree of uncertainty regarding the precise location of the ocean edge. The importance of each component of the ensemble approach is demonstrated in the experiments presented in Section VI-G.

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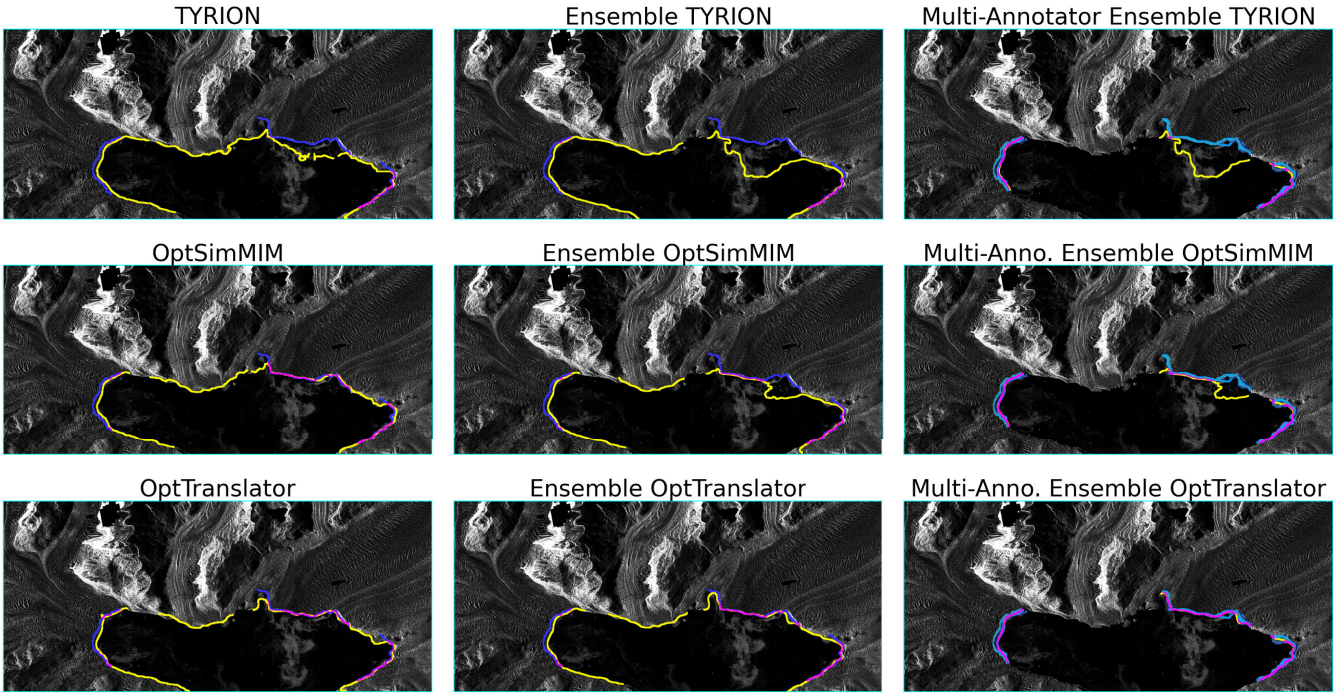


Fig. 7. Visualizations of a sample image of Columbia glacier taken by TanDEM-X (TDX) on October 15, 2014, for TYRION (setup 2) and TYRION with the two pretraining approaches—OptSimMIM and OptTranslator—(setup 4), their corresponding ensembles, and postprocessed ensembles for the multiannotator study. Yellow represents the prediction, blue is used for the ground-truth front, different shades of blue for all ten annotators of the multiannotator study, and pink represents a perfect match between prediction and ground truth/annotator. The bounding box is given by turquoise. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model.

TABLE II

ABLATION STUDY ON THE COMPONENTS OF THE ENSEMBLE APPROACH (SETUP 4). THE THROUGHPUT DENOTES THE TIME OF THE MODEL INCLUDING POSTPROCESSING FOR THE ENTIRE TEST SET DIVIDED BY THE NUMBER OF IMAGES IN THE TEST SET. ALL MODELS WERE INITIALIZED WITH IMAGENET WEIGHTS. THE MDE IS GIVEN IN METERS. A BOLD FONT INDICATES THE BEST VALUE WITHIN A COLUMN

Model	*Pretraining*	*Ensemble*	*Overlap*	*Test-time Aug.*	↑ *Throughput [image/min]*	↓ *MDE*
TYRION	OptSimMIM				11.55	300
	OptTranslator				**11.61**	293
	OptSimMIM	✓			1.00	263
	OptTranslator	✓			0.94	278
	OptSimMIM	✓	✓		0.49	259
	OptTranslator	✓	✓		0.50	262
	OptSimMIM	✓		✓	0.24	242
	OptTranslator	✓		✓	0.23	248
	OptSimMIM	✓	✓	✓	0.13	**238**
	OptTranslator	✓	✓	✓	0.13	246

F. Multiannotator Study

Finally, Table I presents the results of the multiannotator study. The TYRION ensemble model pretrained with SSL4SAR–OptTranslator (setup 4) achieves an MDE of 75 m on the entire test set, which is only 37 m more than the inter-human MDE of 38 m. Predictions for ENVISAT images exhibit the highest MDEs, with a mean deviation from the averaged human annotators' MDE of 325 and 134 m, depending on the pretraining method applied (setup 4). In contrast, TYRION pretrained with SSL4SAR–OptTranslator (setup 4) even surpasses human performance on PALSAR images, achieving an MDE of 100 m compared to 129 m for humans. The results of the multiannotator study are represented graphically in Fig. 9, showing the performance improvements achieved by the novel architecture, the multimodal pretraining techniques, and the ensemble approach. The right side of Fig. 7 illustrates how close the model predictions are to the human annotations, while minor misinterpretations still leave room for improvement.

G. Ablation Studies

We perform an ablation study on the components of our ensemble approach (setup 4), i.e., we remove the test-time augmentation, additional overlap, and the ensembling itself one by one and check the influence on the performance metrics. The results are given in Table II. While MDEs decrease with the additional techniques, the throughput time per image increases. When using ensembling, the model size increases fivefold. In general, test-time augmentations appear to offer greater performance improvements than overlapping patches. For SSL4SAR–OptTranslator, the impact of test-time augmentations even surpasses the performance gains achieved through ensembling. Overlapping patches provide a substantial

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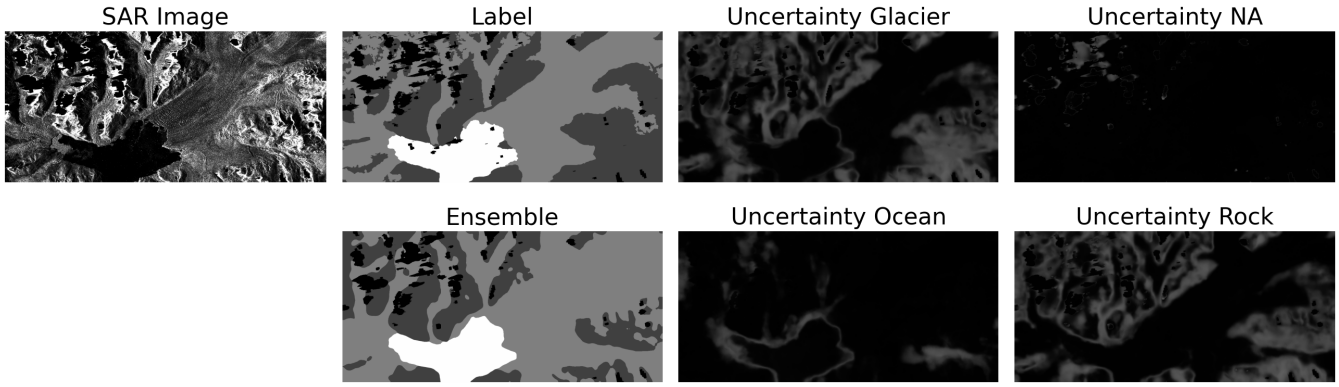


Fig. 8. Sample image of the Columbia glacier captured by S1 on August 14, 2016, with the corresponding zone label, and the ensemble TYRION OptTranslator (setup 4) model's zone segmentation with uncertainty maps per class. The SAR image is shown without contrast enhancement to reflect the raw input provided to the model.

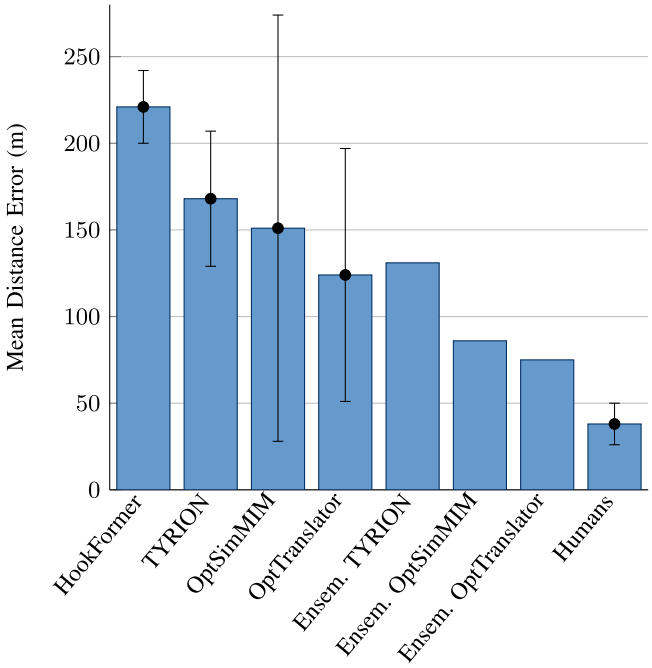


Fig. 9. Overview of the MDEs for the multiannotator study and the results of the postprocessed HookFormer (setup 2), TYRION (setup 2), and TYRION with the two pretraining approaches—OptSimMIM and OptTranslator—(setup 4), as well as the ensembles of the latter three. Confidence intervals are provided for the MDEs of humans and individual models (excluding the ensembles).

performance boost for SSL4SAR–OptTranslator only in the absence of test-time augmentations; otherwise, their contribution is more limited. In the case of SSL4SAR–SimMIM, ensembling yields a greater performance improvement compared to SSL4SAR–OptTranslator. However, overlapping patches and test-time augmentations are less effective for SSL4SAR–SimMIM than for SSL4SAR–OptTranslator. Nevertheless, the overall performance gain achieved using the complete ensemble approach is higher for SSL4SAR–SimMIM than for SSL4SAR–OptTranslator.

H. Discussion

In terms of pretraining, initializing TYRION with ImageNet-pretrained weights without subsequent pretraining on SSL4SAR (setup 2) outperforms SSL pretraining from scratch (setup 3). On the other hand, SSL pretraining starting from ImageNet-pretrained weights (setup 4) outperforms just using ImageNet-pretrained weights (setup 2). This leads us to conclude that supervised pretraining on ImageNet yields more valuable information than SSL pretraining on SSL4SAR. There are two possible reasons for this: 1) the sheer size of ImageNet—over 14 million images versus SSL4SAR's 9562 images and 2) the difference between supervised and self-supervised learning (SSL). Nevertheless, subsequent pretraining of models initialized on ImageNet-pretrained weights with data from the target domain, which are provided by SSL4SAR, still improves the ImageNet-pretrained weights, indicating that some information in SSL4SAR is not present in ImageNet. This is in line with previous works that stress the importance of in-domain pretraining data [14], [15], [16], [17], [18].

The substantial decline in MDE for S1 images resulting from pretraining can be attributed to the fact that SSL4SAR encompasses solely images from S1. If images from other sensors had been incorporated into the dataset, the discrepancy in performance increase between sensors might have been less pronounced. Nevertheless, other sensors also benefit considerably from pretraining on SSL4SAR.

Taking the ensemble leads to an increase in MDE for images captured by ENVISAT. Moreover, ENVISAT images have the highest MDE in the multiannotator study, surpassing all other sensors. One potential explanation for this phenomenon is that the images of this sensor also pose the most difficulties and ambiguities for human annotators, resulting in a high inter-human MDE of 151 m with a substantial standard deviation of 228 m.

VII. Conclusion

This study presents a novel, state-of-the-art deep learning model for delineating glacier calving fronts in SAR imagery. The model is based on an optimized architecture mixing a Swin-Transformer and a residual CNN decoder. An enhanced ensemble approach further improves the results and enables the generation of uncertainty maps. Furthermore, this study proposes two novel multimodal, SSL pretraining strategies for SAR input data that require significantly less optical data than other multimodal remote sensing pretraining approaches.

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To facilitate the pretraining strategies, we introduce a new dataset, SSL4SAR, containing 9562 S1 images of 14 Arctic glaciers and one additional S2 image per glacier. By initializing the ensemble model with ImageNet-pretrained weights and subsequently pretraining it on SSL4SAR, the model achieves an MDE of 238 m on the CaFFe benchmark. The results emphasize the importance of pretraining on an in-domain dataset in addition to supervised pretraining that leverages large, labeled datasets such as ImageNet. With supplementary postprocessing from the multiannotator study, the model achieves an MDE of 75 m when compared to the ground truth from the same study, coming close to the human performance of 38 m.

Code and Data Availability

The code is publicly available on GitHub at https://github.com/Nora-Go/TYRION. The SSL4SAR dataset is publicly available from zenodo at https://doi.org/10.5281/zenodo.14748506. The CAlving Fronts and where to Find thEm (CaFFe) benchmark dataset is publicly available from PANGAEA at https://doi.pangaea.de/10.1594/PANGAEA.940950

Author Contribution

Nora Gourmelon: conceptualization, methodology, software, experiments, statistical analysis, project administration, and writing—original draft preparation. Marcel Dreier: methodology, software, and writing—review and editing. Martin Mayr: validation and writing—review and editing. Thorsten Seehaus: supervision, data curation, and writing—review and editing. Dakota Pyles: writing—review and editing. Matthias Braun, Andreas Maier, and Vincent Christlein: supervision and writing—review and editing.

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